

Progressive Conditional Diffusion Model for Multistage Spectral Restoration of Remote Sensing Image

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Abstract—Due to the high cost and relatively low image quality of hyperspectral sensors, spectral super-resolution seeks to explore the mapping mechanisms between multispectral and hyperspectral images (HSIs), with the goal of reconstructing high-quality HSIs. In recent years, deep learning algorithms have achieved significant success in spectral super-resolution. However, most of these methods compute loss only at the final stage of the network, neglecting the intermediate generative processes, which leads to considerable spectral distortion in the reconstructed images. To address these issues, we propose a progressive conditional diffusion model (PCDM) for multistage spectral restoration. PCDM constructs a channel synthesis module that generates a ground truth set through band synthesis, and designs an image reconstruction module (IRM) to ensure that the synthesized image in the next stage can effectively reconstruct the synthesized features from the previous stage. Multiple conditional diffusion models are then constructed based on the dataset. For each conditional diffusion model, the network parameters of the corresponding IRM are shared with the multispectral image for spectral up-sampling. The spectral-up-sampled multispectral features, combined with the output from the previous diffusion model, serve as a conditional matrix, which is input into the next diffusion model to obtain the final result. Experimental results on both synthetic and real datasets demonstrate that PCDM can effectively reconstruct HSIs, showing robustness and outperforming state-of-the-art methods.

Index Terms—Hyperspectral image (HSI), multispectral image (MSI), multistage diffusion, progressive spectral restoration.

I. INTRODUCTION

HYPERSPECTRAL satellite sensors capture continuous images of the same target with fine spectral resolution, typically less than 20 nm, allowing the detection of subtle

differences between targets [1], [2]. These sensors are widely used in environmental monitoring, resource exploration, smart agriculture, and other fields [3], [4]. However, on the one hand, hyperspectral sensors are complex in design, expensive to manufacture. On the other hand, due to sensor design limitations, hyperspectral images (HSIs) often have low spatial resolution, and the narrow channels have limited ability to absorb reflected energy, making them highly susceptible to environmental influences, such as noise [5], [6]. In contrast to hyperspectral sensors, multispectral sensors have matured in manufacturing technology, are cost-effective and provide higher spatial resolution. However, multispectral images (MSIs) have lower spectral resolution, limiting their ability for fine-grained target identification [7]. To obtain high-quality images at a low cost, spectral super-resolution aims to directly reconstruct HSI with the same spatial resolution from MSI, which has garnered significant attention in recent years [8].

Current spectral super-resolution methods can be broadly categorized into model-driven methods, data-driven methods, and model-data-driven methods [9]. Model-driven methods use fully interpretable mathematical formulas to model the mapping relationship between MSI and HSI [10]. These methods introduce theories such as sparse representation and construct convex objective functions to achieve model convergence through iterative optimization. While these methods offer strong interpretability, they require repeated solving for different datasets, have limited prior information, and are difficult to construct, making nonlinear operations impossible [10].

Data-driven methods, in recent years, have achieved considerable success in image processing and have been widely applied to spectral super-resolution [9]. These methods explore the common nonlinear mappings existing across large datasets, with major network frameworks including convolutional neural network (CNN), generative adversarial networks (GANs), Transformer, and Diffusion [9]. Data-driven methods do not require repeated solving for different datasets, can perform nonlinear operations, and offer significant advantages over model-driven methods. However, they require a large amount of training data [11]. To enhance the interpretability of data-driven methods and improve the performance of model-driven methods, some studies have designed unsupervised model priors to be embedded within data-driven methods, collectively referred to as model-data-driven

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methods. However, these methods still face the challenge of avoiding repetitive model training [10].

Overall, although data-driven methods require a large amount of training data, they maintain a comparative advantage in spectral super-resolution. On the one hand, data-driven models exhibit generalization, do not require repeated training, and provide high accuracy. On the other hand, with advancements in sensor technology, training data can be acquired through various means, making data sources less of a limiting factor. Nonetheless, data-driven methods still face unresolved challenges in constraining intermediate generative processes: While current approaches often employ complex architectures with cascaded subnetworks, most only impose loss constraints at the final output stage, leaving intermediate modules under-regularized. This lack of supervision prolongs training convergence and limits spectral fidelity. Furthermore, even constrained frameworks [e.g., the spectral cascade diffusion model (SCDM) proposed by Chen et al. [12]] may suffer from suboptimal guidance when intermediate sampling strategies (e.g., equidistant GT channel sampling) fail to capture representative features, potentially misdirecting the optimization trajectory and degrading convergence efficiency.

To address the above-mentioned issues, we propose a progressive conditional diffusion model (PCDM) for multistage spectral restoration of remote sensing image. PCDM first constructs multiple conditional diffusion models, wherein several channel synthesis networks with the same architecture are designed to build the GT set for the conditional diffusion models. Specifically, to ensure that the synthesized features obtained during the channel synthesis process are representative, we also design multiple image reconstruction networks with the same architecture. These networks use the features obtained from the channel synthesis process to reconstruct the output features from the channel synthesis network of the previous stage. A loss function is applied to ensure that the features in the current stage are an effective subspace of the features from the previous stage. The channel synthesis features obtained at each stage, which have varying numbers of channels, are input as GT to each conditional diffusion model. Then, we share the parameters of the image reconstruction network with MSI to obtain several datasets that have been preliminarily spectrally up-sampled. These datasets are then used as conditional matrices and input into each diffusion model. Notably, the multispectral reconstruction features at each stage are ultimately combined with the output features of the diffusion model along the channel dimension. After a 1×1 convolution operation for dimensionality reduction, the parameters of the image reconstruction network of the next stage are shared. Finally, after the last stage of the diffusion model, the spectral super-resolution image is obtained. Our main contributions can be summarized as follows.

- 1) We propose a novel PCDM for multistage spectral restoration of remote sensing images. This network ensures the reasonable generation of the GT set and guarantees accurate generation of intermediate features through multistage constraints, achieving high-quality spectral restoration.

- 2) To ensure the reasonable generation of the GT set, we design a global-local channel synthesis and image reconstruction module (IRM). In addition, we design a feature fusion module (FFM) for merging multispectral reconstruction features with the output features of the diffusion model, preventing the loss of valuable information during network computation.
- 3) We validated the performance of PCDM on synthetic and real datasets. Extensive experiments demonstrate that PCDM achieves superior results, outperforming state-of-the-art spectral super-resolution algorithms.

II. RELATED WORK

Model-driven methods leverage mathematical relationships to model the mapping between MSI and HSI [13]. These methods use the inherent properties and statistical information of HSI to learn specific priors for spectral reconstruction. Nguyen et al. [14] employed a radial basis function network to model the mapping relationship between MSI and HSI. Arad and Ben-Shahar [15] proposed a dictionary learning-based spectral super-resolution method using sparse coding. This method constructs a sparse spectral dictionary to obtain hyperspectral atoms, which are then projected onto the multispectral space to establish a mapping between multispectral and hyperspectral atoms, ultimately achieving spectral super-resolution imaging. Aeschbacher et al. [13] improved this method using the A+ algorithm. Specifically, the obtained hyperspectral dictionary serves as an anchor point for nearest-neighbor search to solve for sparse coefficients. Fu et al. [16] learned the spectral dictionary based on non-negative sparse representation and effectively recovered spectral reflectance. Li et al. [17] further incorporated local linear features by embedding the local linearity of spectral data into the dictionary learning process, thus enhancing the super-resolution performance. Subsequently, Geng et al. [18] proposed a spatially constrained dictionary learning method. Considering that HSI can be represented using low-dimensional manifolds, Jia et al. [19] applied manifold learning techniques to reconstruct HSI, though this method requires known illumination conditions and camera parameters, presenting certain limitations. Akhtar and Mian [20] used a Gaussian process to model natural spectra without relying on dictionary learning and combined MSI to reconstruct HSI.

In recent years, data-driven methods have been widely used for spectral super-resolution, primarily involving CNN, GAN, diffusion models, and Transformers [8]. Xiong et al. [21] explored the mapping relationship between MSI and HSI using CNN and proposed HSCNN. The method consists of a spectral up-sampling module and a feature extraction module. The spectral up-sampling module first expands the channels of MSI, and the features are then input into the feature extraction module to reconstruct the HSI. Afterward, Xiong et al. [21] introduced the HSCNN+ network. In this version, HSCNN-U replaces the spectral up-sampling module with convolutional layers, HSCNN-R substitutes residual blocks for standard convolutional layers, and HSCNN-D utilizes an expanded fusion

method to further enhance the super-resolution performance. Given the superior performance of U-Net in image segmentation tasks, which effectively preserves spatial information, Stiebel et al. [22] used a U-Net network without pooling layers to reconstruct HSI. Yan et al. [23] proposed a multiscale U-Net network, which better encodes both local and nonlocal information in MSI. Banerjee and Palrecha [24] replaced the encoder in U-Net with a ResNet using the Mish activation function, designing feature reconstruction loss, style reconstruction loss, and pixel loss to achieve precise spectral super-resolution.

AlvarezGila et al. [25] were the first to apply GAN to spectral super-resolution, achieving promising spectral recovery results. Liu and Zhao [26] improved this approach by incorporating a feature pyramid module into the adversarial network to establish correlations between local and global features, along with dilated convolutions and attention mechanisms to further enhance network performance. Rangnekar et al. [27] used a UNet based generator network and adversarial training to learn the spectral mapping relationship between multispectral and hyperspectral aerial imagery.

Considering the importance of global features in spectral recovery, Transformers have recently shown significant success in image processing. He et al. [28] proposed a dense spectral transformer (DsTer) with ResNet to achieve spectral super-resolution. This method combines Transformers with ResNet, enabling long- and short-range modeling of the image and extraction of global and local features, while dense connections help the model utilize multilevel Transformer features. Liao et al. [29] introduced a lightweight network with convolutional spectral self-attention (TCSSA), which includes a CNN-Transformer encoder, CNN-Transformer decoder, and a convolutional spectral self-attention block. Li et al. [30] cascaded a spectral learning-based Transformer network (SLTN) composed of a spectral response function (SRF)-guided multiscale feature extraction module (MFEM) and spectral nonlinear mapping learning module (NMLM) for image spectral super-resolution. NMLM gradually learns spatial and spectral information from remote sensing images using different blocks, while MFEM addresses atmospheric effects. To enhance the spectral dimension, SRF is integrated into MFEM to guide training.

Given the excellent performance of diffusion models in image generation tasks, Chen et al. [12] proposed a SCDM, which cascades diffusion models along the spectral dimension to gradually refine spectral trends and enrich pixel spectral information. To better harness the potential of diffusion models in spectral super-resolution, the image conditioned mixing guidance was designed to enhance the guidance of image conditions, and progressive dynamic truncation was introduced to limit cumulative errors during the sampling process.

Furthermore, recent model-data-driven approaches have integrated physical models with deep learning for spectral super-resolution. These approaches transform the super-resolution problem into an optimization task and utilize optimization theory to solve the problem, endowing the models with physical interpretability. He et al. [31] proposed an optimization-driven network with deep spatial spectral priors, using SRFs to guide the CNN in initializing spectral band groups. They introduced a

deep spatial spectral prior model for image prior mapping and employed a channel attention mechanism to ensure parameter self-learning, achieving excellent reconstruction performance. Zhu et al. [32] proposed a novel end-to-end learning framework, initially describing the spectral super-resolution problem using imaging models and gradient descent algorithms, and later simulating the correction of the gradient descent process using a lightweight neural network with a multistage structure. They also introduced spectral zero-mean normalization to effectively extract spatial-spectral features. Lin et al. [33] implemented spectral super-resolution based on the ADMM-Adam framework, demonstrating good performance under small sample conditions. Dian et al. [34] introduced a model-guided cross-fusion network, which decomposes the spectral super-resolution network into HSI prior learning (HPL) module and an imaging model guide (IMG) module. The HPL module consists of two distinct subnetworks, effectively learning complex spatial and spectral priors of HSI, while the IMG module solves a strongly convex optimization problem, adaptively optimizing and merging features extracted from the HPL module to achieve excellent results with fewer parameters.

III. METHODOLOGICAL

A. Preliminaries

In image processing, conditional diffusion models are based on the Markov process, where each step of noise injection and denoising depends on the state of the previous step [35]. In the forward diffusion process, the image begins from the original image and gradually undergoes noise addition through multiple steps, with noise addition of each step depending on the image state of the previous step, thereby forming a Markov chain. By the final step, the image has almost completely turned into noise. In the reverse denoising process, the model learns how to reverse this process, gradually removing the noise and recovering the original image, generating an image conditioned on the given information. Each step of denoising also depends on the state of the previous step, forming another Markov process.

Specifically, Given the original data $\mathbf{X}_0$ and the time step t , a Gaussian distribution transformation is obtained with the mean $(\bar{\alpha}_t)^{1/2}\mathbf{X}_0$ and the variance $1 - \bar{\alpha}_t$ as

$$\begin{aligned} q(\mathbf{X}_{1:T}|\mathbf{X}_0) &= \prod_{t=1}^T q(\mathbf{X}_t|\mathbf{X}_{t-1}) \\ &= \mathcal{N}(\mathbf{X}_t; \sqrt{\bar{\alpha}_t}\mathbf{X}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \end{aligned} \quad (1)$$

where $\mathcal{N}$ represents the normal distribution, $\mathbf{I}$ is the identity matrix, $\mathbf{X}_t$ is the image at the t th step, and $\mathbf{X}_{t-1}$ is the image at the $t-1$ th step. $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ is the cumulative noise coefficient, and $\alpha_t = 1 - \beta_t$ represents the signal retention coefficient at each step. The final formula for adding Gaussian noise is obtained using the reparameterization technique as

$$\mathbf{X}_t = \sqrt{\bar{\alpha}_t} \mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \epsilon \in \mathcal{N}(0, 1). \quad (2)$$

The reverse diffusion process restores the clean image from Gaussian noise by employing the same functional form as (1) to reverse the predefined diffusion. The backward process remains

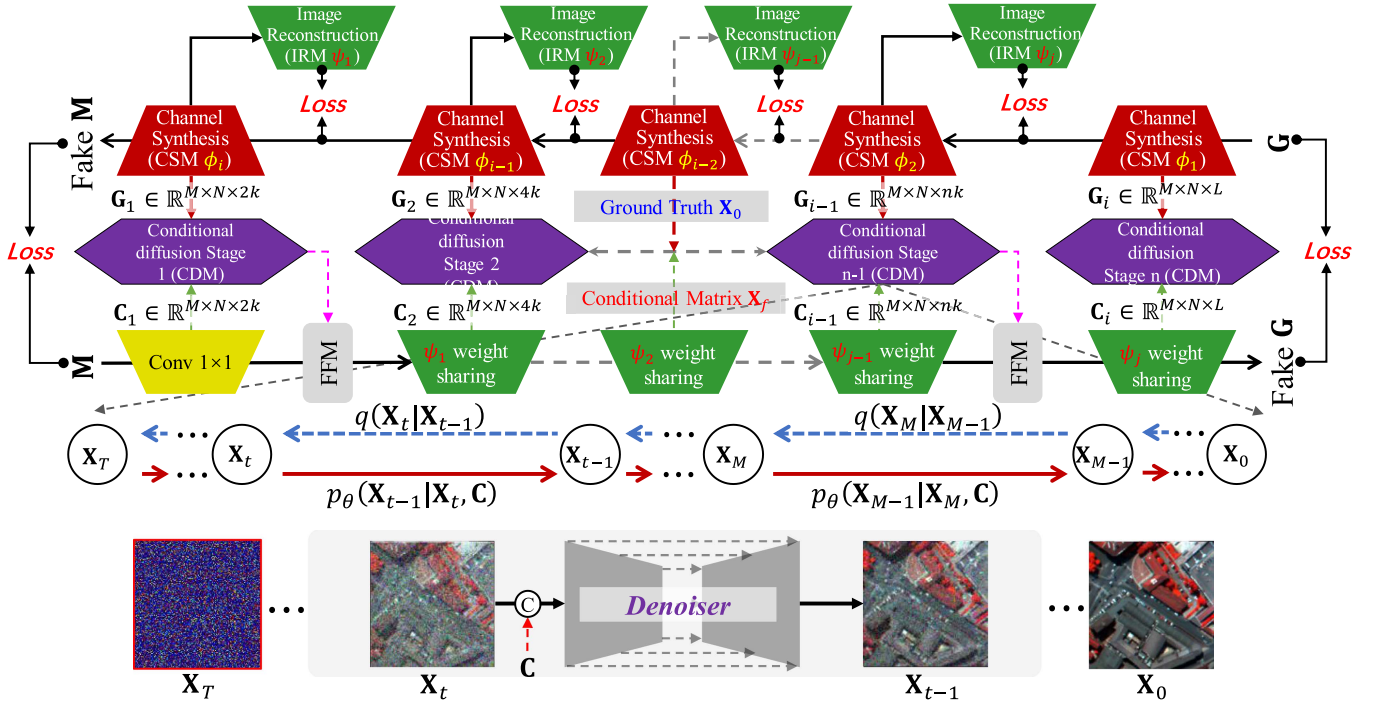


Fig. 1. Progressive conditional diffusion model.

a Markov chain, and eliminates the added Gaussian noise. The conditional information C is incorporated into the inverse diffusion process, which is

$$p_\theta(X_{0:T}) = \prod_{t=1}^T p_\theta(X_{t-1}|X_t, C) = \mathcal{N}\left(X_{t-1}; \mu_\theta(X_t, C, t), \sum_{\theta} (X_t, C, t)\right) \quad (3)$$

where μ_θ is the noise assessment network. The output of the network provides the estimated noise image $\mu_\theta(X_t, C, t)$, which is used to predict the recovered image iteratively through the inverse diffusion as

$$X_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{X_t - \sqrt{1 - \bar{\alpha}_t} \cdot \mu_\theta(X_t, C, t)}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1 - \bar{\alpha}_{t-1}} \cdot \mu_\theta(X_t, C, t). \quad (4)$$

B. Overview of PCDM

Considering that current methods are unable to effectively constrain the intermediate generative process of spectral super-resolution for MSI, leading to poor network convergence and low performance, we propose PCDM for multistage spectral restoration, as shown in Fig. 1. PCDM first explores both the global and local features of GT, and constructs multiple channel synthesis modules (CSMs) to progressively extract subspaces from GT, resulting in GT dataset $G = [G_1, G_2, \dots, G_i]$. It is important to note that we assume GT at the smallest scale should

be consistent with MSI. This can be expressed as

$$G_i = \phi_{i-1}(G_{i-1}) \quad (5)$$

where ϕ represents the network parameters of CSM, and i denotes the i th CSM. However, these synthesized subspaces do not necessarily represent the original GT, and a good subspace should contain the majority of the information from the original image, enabling accurate reconstruction. Therefore, to ensure the validity of the generated G , we design multiple IRMs, using G_i as input to reconstruct G_{i-1} . The effectiveness of G is ensured through a loss function, and the output of the final IRM should be consistent with the original GT. This process can be expressed as

$$G_{i-1} = \psi_j(G_i) \quad (6)$$

where ψ represents the network parameters of IRM, and j denotes the j th IRM. To obtain the conditional matrix set, PCDM shares the parameters of ψ_j with MSI, forming a channel-wise progressive expansion process and resulting in the conditional matrix set $C = [C_1, C_2, \dots, C_i]$, where $C_1 = f_1(M)$, $C_2 = \psi_1(C_1)$, $C_i = \psi_j(C_{i-1})$, M represents MSI, f_1 refers to the 1×1 convolution operation. The rationale is that we designed the CSM to synthesize channels of the ground truth, where the synthesis results of the final stage are computed with the MSI for loss calculation, constituting an HSI-to-MSI transformation process. Meanwhile, to ensure effective information retention in intermediate synthetic images during this conversion, we constructed an IRM that utilizes synthesis results from subsequent stages to reconstruct those from preceding stages, representing an MSI-to-HSI inverse process. Therefore, sharing the IRM parameters with the MSI essentially implements a preliminary

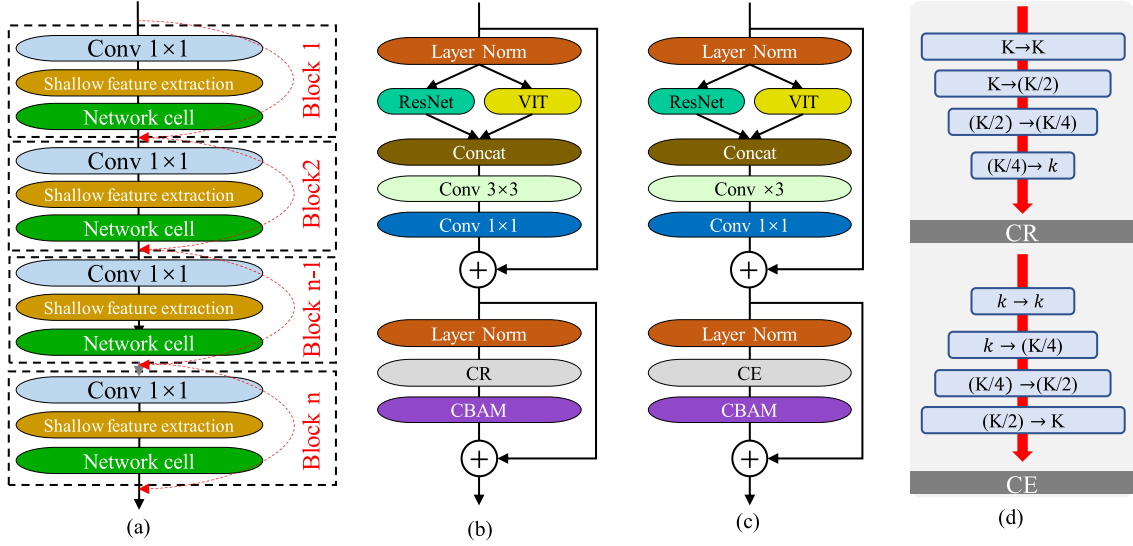


Fig. 2. CSM and IRM. (a) CSM/IRM. (b) CSM Cell. (c) IRM Cell. (d) CR and CE.

spectral super-resolution procedure, where these preliminary super-resolved results serve as conditional matrices to guide the generation of more accurate outcomes in the conditional diffusion model. Ultimately, we construct i conditional diffusion modules (CDMs), using $\mathbf{G}_i$ and $\mathbf{C}_i$ as input data for computation. It is important to note that in the first CDM, we only use $\mathbf{G}_i$ and $\mathbf{C}_i$, and the formula can be defined as

$$\mathbf{O}_{\text{cd}}^1 = \mathcal{T}_1(\mathbf{G}_1, \psi_1(\mathbf{M}_1)) \quad (7)$$

where $\mathcal{T}$ represents CDM, and $\mathbf{O}_{\text{cd}}^1$ denotes the output of the first $\mathcal{T}$. In the second to i th CDMs, PCDM constructs a FFM, which merges the output of the previous CDM with $\mathbf{C}_{i-1}$, and shares the parameters ψ_j to obtain $\mathbf{C}_i$. This can be expressed as

$$\mathbf{O}_{\text{cd}}^i = \mathcal{T}_i(\mathbf{G}_i, \psi_{j-1}(\mathcal{T}_{i-1}(\mathbf{G}_{i-1}, \mathbf{C}_{i-1})), \psi_{j-2}(\mathbf{M}_{i-1})). \quad (8)$$

C. Network Architecture

The proposed PCDM primarily consists of CSM for channel synthesis, IRM for image reconstruction, CDM for channel reconstruction, FFM for feature fusion.

1) *CSM and IRM*: As shown in Fig. 2, both the CSM and IRM share the same backbone network structure. Each network is composed of four blocks, with each block containing a 1×1 convolution operation, shallow feature extraction, and three network cells. It is important to note that, to prevent gradient vanishing, skip connections are also employed between each block to facilitate feature transmission. For each block, the formulation can be expressed as

$$\mathbf{O}_{\text{ci}} = \text{nc}(\text{sfe}(f_1(\mathbf{Z}))) + \mathbf{Z} \quad (9)$$

where the output of the CSM and IRM is given by $\mathbf{O}_{\text{ci}} = [\mathbf{O}_c, \mathbf{O}_i]$, $\mathbf{O}_c$ represents the output features from the CSM, and $\mathbf{O}_i$ denotes the output features from the IRM. $\mathbf{Z}$ is the input data, f_1 refers to the 1×1 convolution operation, and sfe stands

Algorithm: Training PCDM.

Input data: $\mathbf{M}$ and $\mathbf{G}$

Output results: Network parameters

Data processing: Normalize $\mathbf{M}$ and $\mathbf{G}$ to 0-255

for $i = 1$: epoch

Stage-1: Calculate G and C Equations (5)–(6)

- Perform global-local modeling on the channel relationships of $\mathbf{G}$ and conduct channel synthesis to obtain $G = [\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_i]$.
- Reconstruct $\mathbf{G}$ through inverse global-local modeling based on the results from step a.
- Share the parameters from step b to $f_1(\mathbf{M})$ to obtain $C = [\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_i]$.

Stage-2: Calculate $\mathbf{O}_{\text{cd}}^i$ Equations (7)–(8)

- Simulate the noisy Gaussian diffusion process to construct T-step degraded features for $\mathbf{G}_1$
- Infer noise information via a $\mathbf{C}_1$ -conditioned reverse Markov process.
- Generate $\mathbf{C}_2$ using the result of step e, and perform the operation of step d using $\mathbf{G}_2$, and perform the operation of step e using $\mathbf{C}_2$ until the data of the i th scale is calculated.

Loss calculation: Optimize network parameters

End

for shallow feature extraction. $\text{nc} = [\text{CSM cell}, \text{IRM cell}]$, CSM cell and IRM cell are the respective components of the CSM and IRM.

For shallow feature extraction, both CSM and IRM employ a ResNet with three convolutional layers for the preliminary extraction of features. However, shallow features alone often fail to provide sufficient context or global semantics in image

processing tasks, whereas deep features help the model make more accurate judgments. Therefore, we utilize network cells to explore the deeper expressions of information based on the shallow features. Regarding the network cells, numerous studies have shown that in image processing, it is crucial to consider not only the local similarity of the target but also the global correlations of the information. Extracting both local and global information significantly enhances image processing performance. In addition, CNN have been proven effective in modeling short-range dependencies between targets, while transformers capture long-range relationships between information. Thus, we combine ResNet and ViT to construct the CSM cell and IRM cell. Notably, both the CSM cell and IRM cell share the same network structure, each consisting of two stages. In the first stage, the cell applies layer normalization to the input data, and the processed features are separately passed through ResNet and ViT to capture local and global information. Subsequently, the local and global information is concatenated along the channel dimension, then processed through 3×3 and 1×1 convolution operations, and added back to the original input data.

$$\hat{\mathbf{Z}} = \ln(\mathbf{Z}) \quad (10)$$

$$\mathbf{O}_{\text{cell}}^{s1} = f_1 \left(f_3 \left(\text{cat} \left(R \left(\hat{\mathbf{Z}} \right), V \left(\hat{\mathbf{Z}} \right) \right) \right) + \mathbf{Z} \quad (11)$$

where the output of the first-stage cell is denoted as $\mathbf{O}_{\text{cell}}^{s1}$, $\ln$ refers to the layer normalization operation, f_3 is the 3×3 convolution operation, and cat represents the concatenation operation along the channel dimension. R refers to the residual network, and V denotes the vision transformer network. In the second stage, the output from the first stage undergoes layer normalization, followed by the application of a channel adjustment module that either synthesizes or expands the channels of the output features. Afterward, the convolutional block attention module (CBAM) is used to reoptimize the channel-adjusted features, which are then added to the first-stage output features to obtain the final output features

$$\mathbf{O}_{\text{cell}}^{s2} = \text{an} \left(\text{ca} \left(\ln \left(\mathbf{O}_{\text{cell}}^{s1} \right) \right) \right) + \mathbf{O}_{\text{cell}}^{s1} \quad (12)$$

where the output of the second-stage cell is denoted as $\mathbf{O}_{\text{cell}}^{s2}$, $\text{ca} = [\text{CR}, \text{CE}]$, where CR represents the CSM and CE represents the channel expansion module. an refers to the CBAM network.

In each cell, the ResNet uses 3×3 convolution operations, which effectively extract local features from the image while avoiding gradient vanishing. It is primarily composed of three residual blocks, and each residual block consists of five convolutional layers. In addition, ViT overcomes the local window operation mechanism of ResNet by utilizing the self-attention mechanism for image feature extraction.

As shown in Fig. 2(d), the channel reconstruction (CR) and channel expansion (CE) modules have symmetric network structures, each consisting of four layers of 1×1 convolution operation. The number of channels is adjusted for either reduction or expansion based on specific requirements.

2) *Conditional Diffusion Module*: PCDM decomposes MSI and GT into four datasets $\{[\mathbf{G}_1, \mathbf{C}_1], [\mathbf{G}_2, \mathbf{C}_2], [\mathbf{G}_3, \mathbf{C}_3], [\mathbf{G}_4, \mathbf{C}_4]\}$ through CSM and IRM, and CDM to

progressively perform spectral super-resolution of MSI over multiple stages. For the CDM, the denoising network is crucial, and numerous studies have employed UNet as the denoising model for CDM. This choice is due to its powerful feature extraction capability and the advantage of multiscale information fusion, enabling effective integration of low-level detail and high-level semantic features via skip connections. This helps in recovering image details while preserving structural integrity.

In our experiments, we also utilized the UNet network as the denoising model. In the encoder part, the network gradually reduces the spatial resolution of the image and extracts high-level features through multiple convolutional and pooling layers. Each convolutional layer typically uses a 3×3 kernel with a stride of 1, and with each layer, the output channel number increases incrementally. Starting from an input image with k channels, the first layer outputs 64 channels, the second layer outputs 128 channels, and so on, until the maximum output of 512 channels is reached. The pooling layer employs 2×2 max pooling with a stride of 2, reducing the spatial size of the feature map. In the bottleneck part, the network extracts the deepest features. In the decoder part, the network progressively restores the spatial resolution using deconvolutional layers, while skip connections combine features from the corresponding layers of the encoder and decoder, thus maintaining image details. Each deconvolutional layer uses a 3×3 kernel with a stride of 1, and the output channel number gradually decreases, matching the features from the encoder part. The final output denoised image has the same dimensions and number of channels as the input image.

3) *Feature Fusion Module*: The connectivity of the output features from the multistage CDM is crucial. In the experiment, we designed FFM to integrate the output from the CDM of previous stage with multispectral features, as shown in Fig. 3. The fused features were then input into IRM of the next stage for further processing. Considering the similarity between the output features $\mathbf{X}_{\text{dcm}}$ of the CDM of same stage and the multispectral features $\mathbf{X}_{\text{msi}}$, there is inherent information redundancy. This redundancy may hinder network inference. To mitigate redundant information among features, we first subtracted $\mathbf{X}_{\text{dcm}}$ from $\mathbf{X}_{\text{msi}}$ to obtain the differential features $\mathbf{E}$ between the images. We then applied a sigmoid activation function to analyze the distribution characteristics of $\mathbf{E}$ and derived the attention matrix $\mathbf{A}$. Subsequently, we multiplied $\mathbf{A}$ with the 1×1 convolutional output of $\mathbf{X}_{\text{dcm}}$ and $\mathbf{X}_{\text{msi}}$ to obtain the unique interfeature information of $\mathbf{X}_{\text{dcm}}$ and $\mathbf{X}_{\text{msi}}$. Finally, skip connections were used to add the attention information back to $\mathbf{X}_{\text{dcm}}$ and $\mathbf{X}_{\text{msi}}$ independently of $\mathbf{E}$, thus obtaining the refined features of $\mathbf{X}_{\text{dcm}}$ and $\mathbf{X}_{\text{msi}}$

$$\mathbf{A} = \delta(f_3(f_1(\mathbf{X}_{\text{dcm}} - \mathbf{X}_{\text{msi}}))) \quad (13)$$

$$\mathbf{X}_{\text{dcm}}^e = \mathbf{X}_{\text{dcm}} + f_1(\mathbf{X}_{\text{dcm}}) \times \mathbf{A} \quad (14)$$

$$\mathbf{X}_{\text{msi}}^e = \mathbf{X}_{\text{msi}} + f_1(\mathbf{X}_{\text{msi}}) \times \mathbf{A} \quad (15)$$

where δ denotes the sigmoid activation function, and $\mathbf{X}_{\text{dcm}}^e$ and $\mathbf{X}_{\text{msi}}^e$ are the output features. After eliminating the redundant

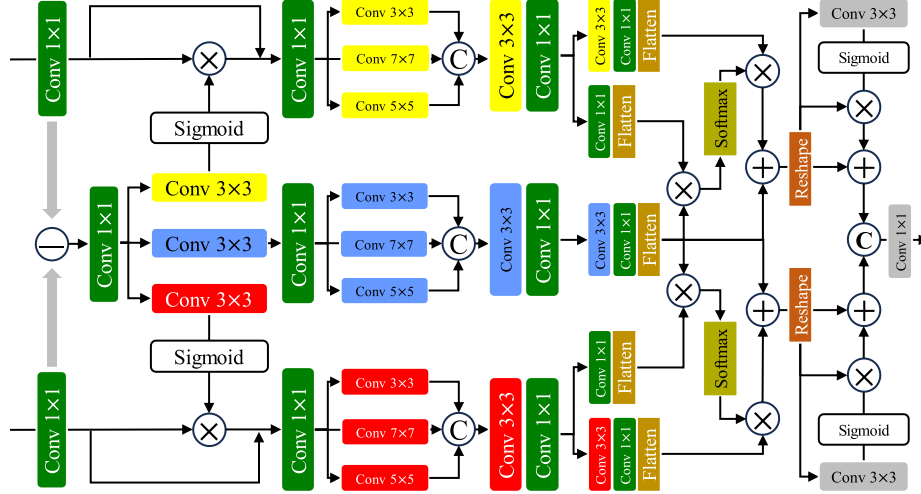


Fig. 3. Feature fusion module.

information between the features, we performed multiscale feature extraction and fusion on $\mathbf{X}_{\text{dcm}}^e$, $\mathbf{X}_{\text{msi}}^e$, and $\mathbf{E}$, respectively

$$\mathbf{X}_{\text{dcm}}^{\text{me}} = f_1(f_3(\mathcal{M}_d(\mathbf{X}_{\text{dcm}}^e))) \quad (16)$$

$$\mathbf{X}_{\text{msi}}^{\text{me}} = f_1(f_3(\mathcal{M}_m(\mathbf{X}_{\text{msi}}^e))) \quad (17)$$

$$\mathbf{E}_{\text{me}} = f_1(f_3(\mathcal{M}_e(\mathbf{E}))) \quad (18)$$

where $\mathbf{X}_{\text{dcm}}^{\text{me}}$, $\mathbf{X}_{\text{msi}}^{\text{me}}$, and $\mathbf{E}_{\text{me}}$ represent the output features, and $\mathcal{M}$ denotes the multiscale feature extraction network, which includes convolution operations with kernel sizes of 3×3 , 5×5 , and 7×7 . To achieve comprehensive information fusion, a cross-attention mechanism is employed to fuse $\mathbf{X}_{\text{dcm}}^{\text{me}}$ with $\mathbf{E}_{\text{me}}$, and $\mathbf{X}_{\text{msi}}^{\text{me}}$ with $\mathbf{E}_{\text{me}}$ respectively. This can be expressed as

$$\begin{aligned} \mathbf{Q}_d &= fl(f_1(\mathbf{X}_{\text{dcm}}^{\text{me}})), \mathbf{K}_d = fl(f_1(f_3(\mathbf{E}_{\text{me}}))) \\ \mathbf{V}_d &= fl(f_1(f_3(\mathbf{X}_{\text{dcm}}^{\text{me}}))) \end{aligned} \quad (19)$$

$$\mathbf{X}_{\text{dcm}}^A = \text{softmax}(\mathbf{Q}_d \mathbf{K}_d^T) \mathbf{V}_d + \mathbf{E}_{\text{me}} \quad (20)$$

$$\begin{aligned} \mathbf{Q}_m &= fl(f_1(\mathbf{X}_{\text{msi}}^{\text{me}})), \mathbf{K}_m = fl(f_1(f_3(\mathbf{E}_{\text{me}}))), \\ \mathbf{V}_m &= fl(f_1(f_3(\mathbf{X}_{\text{msi}}^{\text{me}}))) \end{aligned} \quad (21)$$

$$\mathbf{X}_{\text{msi}}^A = \text{softmax}(\mathbf{Q}_m \mathbf{K}_m^T) \mathbf{V}_m + \mathbf{E}_{\text{me}} \quad (22)$$

where $\mathbf{X}_{\text{dcm}}^A$ and $\mathbf{X}_{\text{msi}}^A$ represent the output features. Finally, $\mathbf{X}_{\text{dcm}}^A$ and $\mathbf{X}_{\text{msi}}^A$ are reoptimized, and after channel dimension concatenation, the resulting features undergo dimensionality reduction through a 1×1 convolution operation to obtain the final output features

$$\mathbf{X}_{\text{dcm}}^f = \mathbf{X}_{\text{dcm}}^A + \mathbf{X}_{\text{dcm}}^A \times \delta(f_3(\mathbf{X}_{\text{dcm}}^A)) \quad (23)$$

$$\mathbf{X}_{\text{msi}}^f = \mathbf{X}_{\text{msi}}^A + \mathbf{X}_{\text{msi}}^A \times \delta(f_3(\mathbf{X}_{\text{msi}}^A)) \quad (24)$$

$$\mathbf{O}_{\text{fm}} = f_1(\text{cat}(\mathbf{X}_{\text{dcm}}^f, \mathbf{X}_{\text{msi}}^f)) \quad (25)$$

where $\mathbf{O}_{\text{fm}}$ represents the output features of the FFM.

D. Training Details

To train PCDM, we constructed the image reconstruction loss $\mathcal{L}_{ir}$ and the diffusion loss $\mathcal{L}_d$. It is important to note that $\mathcal{L}_{ir} = [\mathcal{L}_{ir}^1, \mathcal{L}_{ir}^2, \mathcal{L}_{ir}^3, \mathcal{L}_{ir}^4]$ and $\mathcal{L}_d = [\mathcal{L}_d^1, \mathcal{L}_d^2, \mathcal{L}_d^3, \mathcal{L}_d^4]$. In the experiments, both $\mathcal{L}_{ir}$ and $\mathcal{L}_d$ are computed based on the MSE loss function, with $\mathcal{L}_{ir}$ defined as

$$\begin{aligned} \mathcal{L}_{ir} &= \sum_{i=1}^{n=4} \mathcal{L}_{ir}^i \\ \text{s.t. } \mathcal{L}_{ir}^i &= \frac{1}{2m} \sum_{j=1}^m \|\text{irm}_i(\mathbf{G}_j^{i-1}) - \mathbf{G}_j^i\|_2^2 \end{aligned} \quad (26)$$

where $n = 4$ represents the number of stages, $\mathcal{L}_{ir}^i$ denotes the image reconstruction loss at the i th stage, m is the number of pixels, and j represents the j th pixel, irm referring to IRM. In addition, $\mathcal{L}_d$ is defined as follows:

$$\begin{aligned} \mathcal{L}_d &= \sum_{i=1}^{n=4} \mathcal{L}_d^i \\ \text{s.t. } \mathcal{L}_d^i &= \mathbb{E}_{\mathbf{X}_t^i, \mathbf{C}_i, \epsilon, t} [\|\epsilon - \epsilon_\theta(\mathbf{X}_t^i, t, \mathbf{C}_i)\|^2], \mathbf{X}_t^i = \mathbf{G}_t^i \end{aligned} \quad (27)$$

where $\mathcal{L}_d^i$ denotes the diffusion loss at the i th stage, ϵ is the added noise, $\epsilon_\theta(\mathbf{X}_t^i, t, \mathbf{C}_i)$ is the noise predicted by the model at time step t , with condition $\mathbf{C}_i$, and $\mathbb{E}$ represents the expected value. Finally, the overall loss function $\mathcal{L}_{\text{pcdm}}$ is defined as

$$\mathcal{L}_{\text{pcdm}} = \mathcal{L}_{ir} + \mathcal{L}_d. \quad (28)$$

IV. CONCLUSION

In this section, we conducted performance evaluations of the PCDM using both synthetic and real-world datasets, and compared it with seven state-of-the-art methods. In addition, we performed ablation experiments on key components to demonstrate the effectiveness and rationality of the PCDM design.

A. Experimental Datasets

Synthetic Datasets: 1) *Pavia Center Dataset (PaviaC)*: The Pavia dataset is a hyperspectral dataset widely used in remote sensing image analysis. This dataset is sourced from the city of Pavia in northern Italy and was collected by the German airborne Reflective Optical Spectral Imager (ROSIS-03) in 2003. It contains 103 bands. In the experiment, 80 bands were used for experimental verification after removing the bad bands. We obtained MSI with four bands by spectral sampling using the Gaofen (GF)-1 SRF. Ultimately, both HSI and MSI were cropped into 256×256 pixel for network training, with an overlap of 32×32 pixels between adjacent patches. By applying mirroring and rotation, a total of 5136 patches were obtained.

2) *Xiongan Dataset*: The Xiongan Dataset was collected using the Full-Spectral Multimodal Imaging Spectrometer installed within the High-Resolution Special Aviation System, developed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences. The spectral range of this dataset spans from 400 to 1000 nanometers, encompassing 250 spectral bands. The image size is 3750×1580 pixels, with a spatial resolution of 0.5 m. In the experiment, 170 high-quality spectral bands were retained for experimental validation. We obtained MSI with four bands by spectral sampling using the Landsat-4 SRF. Ultimately, both HSI and MSI were cropped into 256×256 pixel for network training, with an overlap of 128×128 pixels between adjacent patches. By applying mirroring and rotation, a total of 7653 patches were obtained.

3) *Chikusei Dataset*: HSI was captured by the Headwall Hyperspec-VNIR-C sensor in Chikusei, Japan, on 29 July 2014 [36]. The dataset contains 128 spectral bands ranging from 343 to 1018 nm, with a size of 2517×2335 pixels and a spatial resolution of 2.5 m, covering both urban and rural areas. We obtained MSI with four bands by spectral sampling using the SPOT SRF. Ultimately, both HSI and MSI were cropped into 256×256 pixel for network training, with an overlap of 64×64 pixels between adjacent patches. By applying mirroring and rotation, a total of 9658 patches were obtained.

Real Datasets: *PRISMA* [37]: PRISMA is a hyperspectral satellite led by the Italian Space Agency (ASI), designed to provide high-precision spectral data for Earth observation. The satellite was successfully launched on 22 March 2019, with a spectral range covering 400–2500 nm, including visible light, near-infrared, and shortwave infrared regions. The spectral resolution is approximately 5 nm in the range of 450–1000 nm and about 10 nm in the 1000–2500 nm range, with a total of 239 spectral bands. The spatial resolution is 30 m, allowing for precise ground target identification, with a swath width of 30 km. PRISMA HSI is widely used in agriculture, forestry, water resource management, and environmental monitoring. In the experiment, these HSI were preprocessed with orthorectification, radiometric calibration, atmospheric correction, and removal of bands with severe water vapor absorption, with 100 bands selected for training. *Landsat-8* [38]: Landsat-8 is an Earth observation satellite launched by NASA on 11 February 2013, with a spectral range covering visible light, near-infrared, and shortwave infrared bands. The spatial resolution is 30 m, with a

swath width of 185 km, making it widely applicable in agriculture, forestry, disaster monitoring, urban planning, and more. In the experiment, six bands with a spatial resolution of 30 m were used. We collected PRISMA-Landsat-8 images from January to December 2023. After cropping, 11 PRISMA-Landsat-8 image datasets with 1200×900 pixels were obtained. These images were further cropped into 256×256 pixel for network training, with an overlap of 64×64 pixels between adjacent patches, yielding a total of 5069 patches after applying mirroring and rotation.

B. Experimental Settings

To evaluate the spectral super-resolution performance of PCDM, six methods were selected for comparative experiments: the traditional method A+ (2017), the CNN-based method CNN-SSR (2021) [39], the GAN-based method GAN-SSR (2017) [25], the Transformer-based methods DsTer (2022) [28] and SLTN (2023) [30], the Diffusion-based method SCDM (2024) [12], and low-rank tensor reconstruction network (LTRN) (2024) [40]. It is noteworthy that the parameter settings for A+ were derived from the original references. For all deep learning-based methods, the network structures and parameter configurations were also taken from the optimal references, and the number of epochs was set to 200 to explore the convergence rates of different methods.

The evaluation metrics used to assess the performance of the different methods include peak signal-to-noise ratio (PSNR), root mean square error (RMSE), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), spectral angle map (SAM), and cross-correlation (CC) [41]. Higher values of PSNR and CC indicate better results, while higher values of RMSE, ERGAS, and SAM suggest poorer performance, with an ideal value of 0.

For training PCDM, the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$ was used, the learning rate was set to $1e-4$, and the batch size was set to 4. In addition, PCDM was trained on a Windows 11 computer with 128 GB of CPU memory and a $1 \times$ GeForce GTX 4090D GPU.

C. Spectral Super-Resolution With Synthetic Datasets

1) *PaviaC Dataset*: Fig. 4 illustrates the spectral super-resolution results on the PaviaC dataset. The first row displays pseudo-color composites generated from bands (25, 15, 5). All methods preserve well-defined spatial details, but the A+ method exhibits color discrepancies relative to the reference image, accompanied by pronounced spectral distortion. In contrast, other methods maintain spectral consistency with the reference.

The second row presents SAM error distributions. The A+ method demonstrates substantial spectral deviations, whereas CNN-SSR, GAN-SSR, DsTer, SLTN, LRTN, and SCDM yield significantly lower errors, with our method achieving minimal spectral reconstruction error.

Band 75 visualization in the third row confirms the alignment of benchmark methods with the reference band. Quantitative analysis of absolute errors (fourth row) reveals that A+ produces the largest deviation, followed by CNN-SSR.

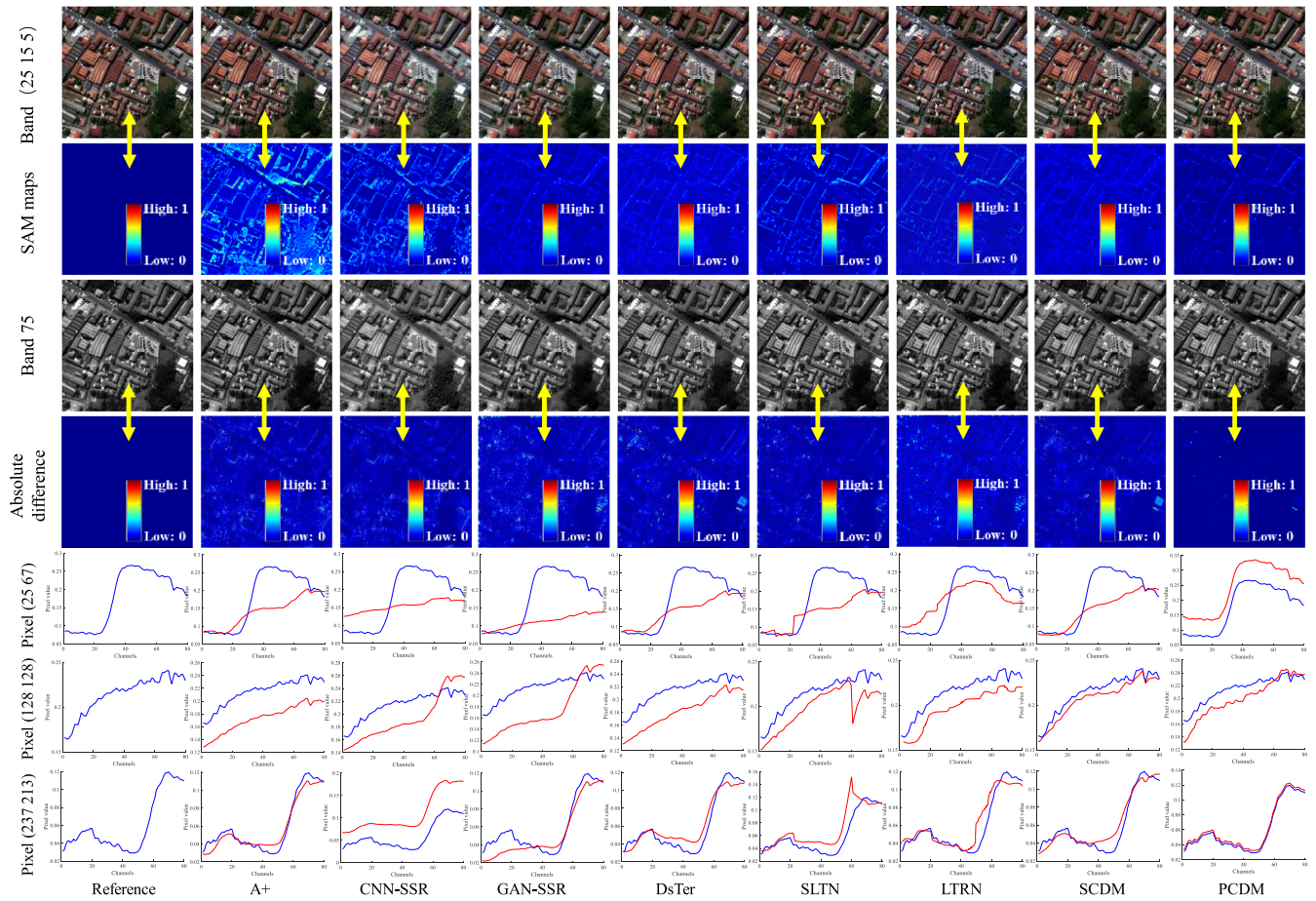


Fig. 4. Experimental results of PaviaC dataset.

Errors in GAN-SSR, DsTer, SLTN, LRTN, and SCDM primarily localize at object boundaries, while our method surpasses all counterparts.

Rows 5–7 display spectral curve reconstructions for representative targets at pixels (25,67), (128,128), and (237,213). The A+ method generates the least accurate reconstructions. CNN-SSR and GAN-SSR approximate reference trends but exhibit unstable fluctuations. DsTer, SLTN, LRTN, and SCDM demonstrate enhanced stability, and our proposed PCDM achieves near-perfect correspondence to reference spectra.

2) *Xiongan Dataset*: Fig. 5 demonstrates the spectral super-resolution results on the Xiongan dataset. The first row presents pseudo-color composites synthesized from bands 45, 25, and 5. All methods maintain clear spatial details. However, the A+ method exhibits significant global color deviations compared to the reference image, while CNN-SSR shows localized discrepancies in bridge and road regions. GAN-SSR, DsTer, SLTN, LRTN, and SCDM display minor color differences concentrated in road areas, with the proposed PCDM achieving the closest spectral fidelity to the reference.

SAM error distributions in the second row highlight severe global spectral distortion in A+. Other benchmark methods primarily show spectral errors in vegetation and roadbed regions. Among them, CNN-SSR yields the highest reconstruction error,

whereas GAN-SSR, DsTer, SLTN, LRTN, and SCDM achieve minimal spectral distortion. Although PCDM retains minor errors in vegetation and roadbed areas, it consistently maintains the lowest distortion overall.

Band 140 visualization in the third row confirms consistent grayscale distributions across all methods except A+. Absolute error maps in the fourth row further quantify performance. A+ produces the largest deviations, while reconstruction errors in GAN-SSR, DsTer, SLTN, LRTN, and SCDM localize to bridge and roadbed structures. PCDM achieves the smallest error.

Spectral curves of three representative targets at pixels (67,32), (187,153), and (216,236) are displayed in rows 5–7. A+ performs poorly across all cases. CNN-SSR aligns with reference trends for pixels (67,32) and (187,153) but fails to reconstruct the spectral curve for (216,236), showing unstable fluctuations. GAN-SSR, DsTer, SLTN, LRTN, and SCDM follow reference trends but exhibit spectral instability, particularly peak-valley oscillations in GAN-SSR, DsTer, and SLTN. SCDM outperforms other benchmarks, and the proposed PCDM achieves near-perfect spectral correspondence with minimal reconstruction errors across all targets.

3) *Chikusei Dataset*: Fig. 6 presents the experimental results of different methods on the Chikusei dataset. The first row displays pseudo-color composites synthesized from bands

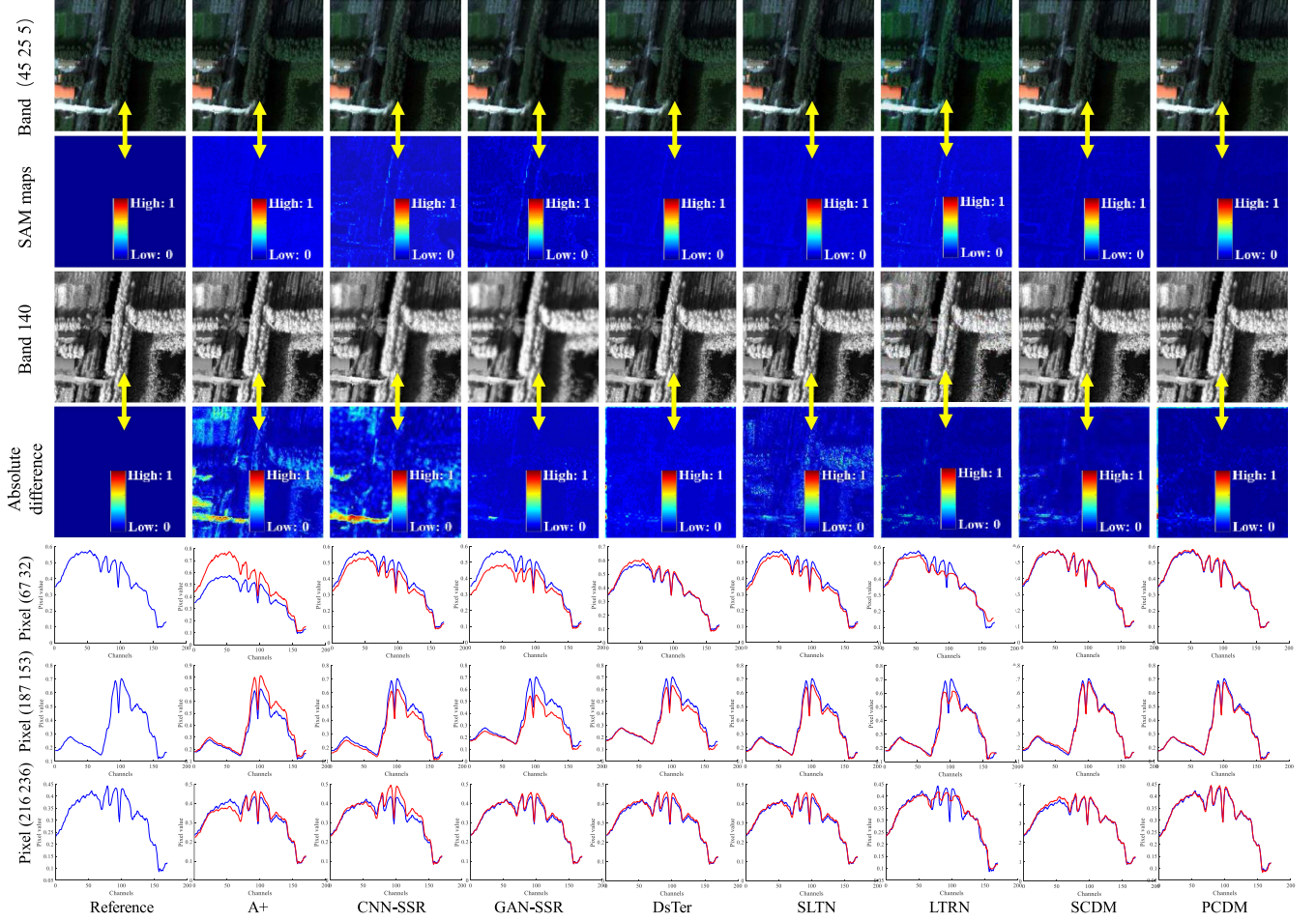


Fig. 5. Experimental results of Xiong'an dataset.

(88, 15, 5). While the A+ method preserves spatial details clearly, it exhibits significant color deviations compared to the reference image. CNN-SSR, GAN-SSR, DsTer, SLTN, and LRTN maintain color distributions consistent with the reference but lose partial spatial details. Both SCDM and the proposed PCDM retain prominent spatial structures while achieving the closest spectral alignment with the reference.

SAM error distributions (second row) reveal severe global spectral distortion in A+. CNN-SSR, GAN-SSR, DsTer, SLTN, LRTN, and SCDM show localized spectral errors in artificial building regions. Although PCDM displays minor spectral reconstruction errors in water bodies, it consistently achieves the lowest overall distortion.

Band 74 visualization (third row) demonstrates inconsistent grayscale distributions in A+, CNN-SSR, and GAN-SSR compared to the reference. In contrast, DsTer, SLTN, LRTN, and SCDM align well with the reference. Absolute error maps (fourth row) confirm significant reconstruction errors in A+ and CNN-SSR, while other methods exhibit smaller errors. PCDM achieves the minimal error overall.

Spectral curves of representative targets at pixels (58,27), (159,121), and (241,237) (rows 5–7) indicate that A+ and CNN-SSR approximate reference trends for (58,27) and

(159,121) but exhibit unstable fluctuations. Both methods fail to reconstruct the spectral curve for (241,237). GAN-SSR, DsTer, SLTN, LRTN, and SCDM follow reference trends but show spectral instability (peak-valley oscillations in GAN-SSR, DsTer, and SLTN). SCDM outperforms all benchmarks, while PCDM achieves optimal spectral reconstruction accuracy.

4) *Quantitative Evaluation Results:* Table I summarizes quantitative evaluations across datasets. On the PaviaC dataset, all methods except A+ achieve PSNR >28, RMSE <0.04, ERGAS <8, CC >0.96, and SSIM >0.88. A+ yields the poorest metrics, while CNN-SSR and GAN-SSR perform comparably. DsTer, SLTN, LRTN, and SCDM surpass other benchmarks, with PCDM attaining the highest scores.

For the Xiong'an dataset, A+ shows the worst performance (PSNR <30, RMSE >0.025, SAM >3.1, SSIM <0.935). SCDM ranks highest among benchmarks, followed by DsTer, SLTN, and LRTN. CNN-SSR and GAN-SSR exceed A+ but underperform relative to other methods. PCDM achieves the best quantitative results.

On the Chikusei dataset, A+ again performs weakest across all metrics. CNN-SSR and GAN-SSR exhibit the lowest PSNR, RMSE, CC, and SSIM, with CNN-SSR showing the highest

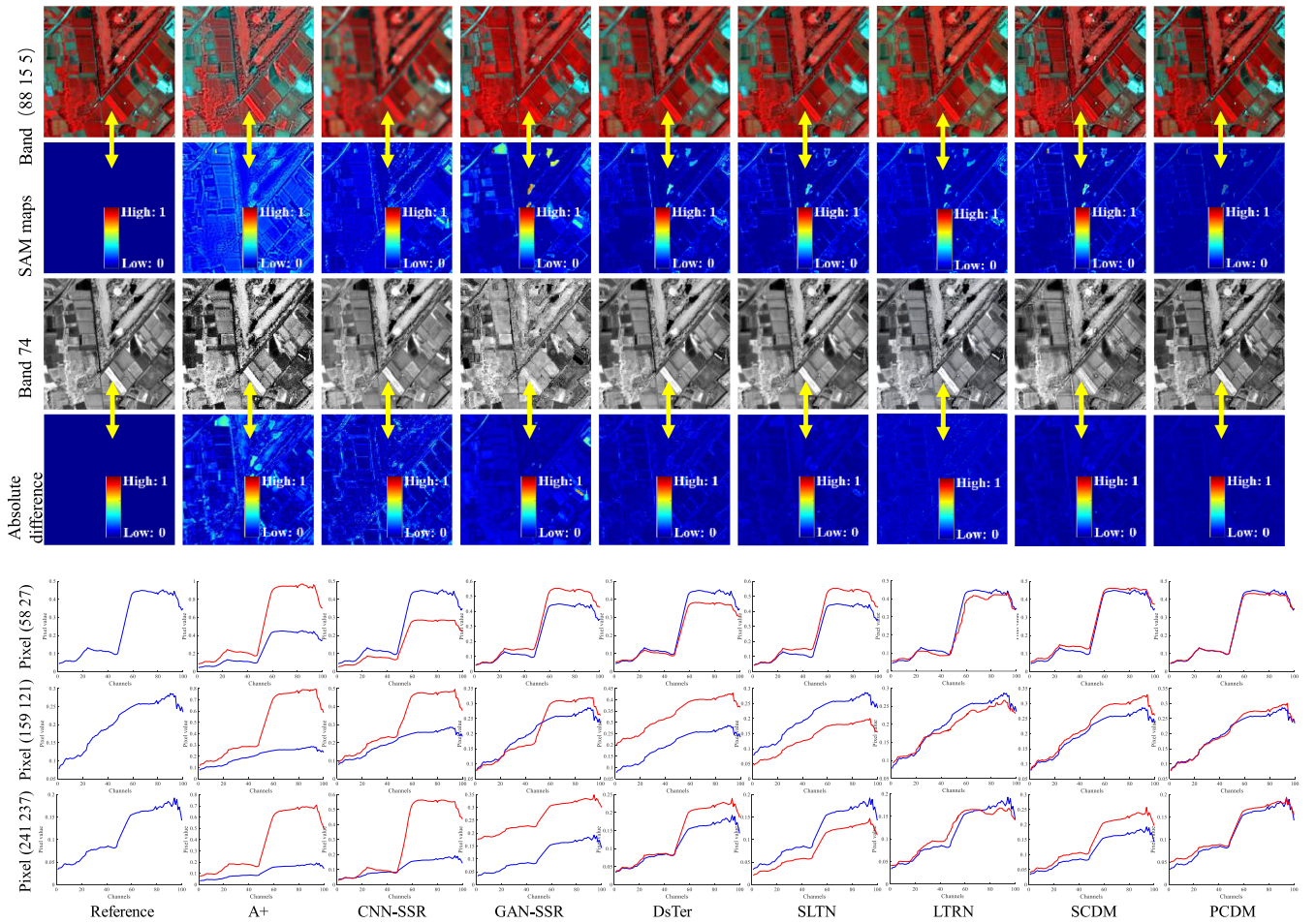


Fig. 6. Experimental results of Chikusei dataset.

TABLE I
QUANTITATIVE EVALUATION EXPERIMENT RESULTS

Methods	A+	CNN-SSR	GAN-SSR	DsTer	SLTN	LRTN	SCDM	PCDM
PaviaC Datasets								
PSNR	21.71	28.20	28.78	34.02	35.17	34.69	35.30	41.84
RMSE	0.089	0.039	0.038	0.021	0.019	0.018	0.018	0.008
ERGAS	9.29	6.05	7.56	3.71	3.14	4.21	3.18	1.60
SAM	8.07	2.22	4.02	2.40	3.55	3.98	2.29	2.11
CC	0.936	0.967	0.977	0.985	0.989	0.991	0.994	0.997
SSIM	0.874	0.882	0.899	0.973	0.966	0.968	0.978	0.989
Xiong'an Datasets								
PSNR	28.50	31.99	31.42	39.06	36.72	37.73	41.25	42.27
RMSE	0.030	0.021	0.024	0.008	0.010	0.008	0.007	0.006
ERGAS	2.61	2.23	2.52	0.85	0.96	1.43	0.68	0.64
SAM	3.30	3.01	2.67	1.02	1.30	1.51	1.00	0.97
CC	0.931	0.986	0.979	0.986	0.985	0.989	0.991	0.998
SSIM	0.934	0.937	0.958	0.981	0.988	0.986	0.990	0.992
Chikusei Datasets								
PSNR	25.63	31.26	30.94	33.68	32.92	33.65	35.16	38.43
RMSE	0.128	0.073	0.101	0.028	0.032	0.027	0.023	0.016
ERGAS	11.82	7.83	7.29	4.85	4.37	4.20	4.33	3.06
SAM	9.34	5.15	4.26	2.77	2.73	2.58	2.66	1.98
CC	0.915	0.913	0.861	0.960	0.964	0.969	0.979	0.989
SSIM	0.646	0.834	0.748	0.910	0.909	0.937	0.956	0.987

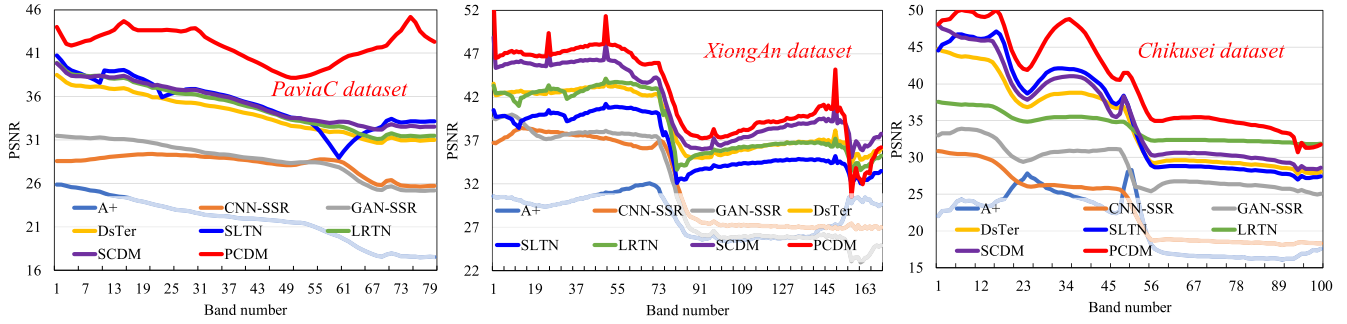


Fig. 7. SSIM for each band using different methods in the synthetic dataset.

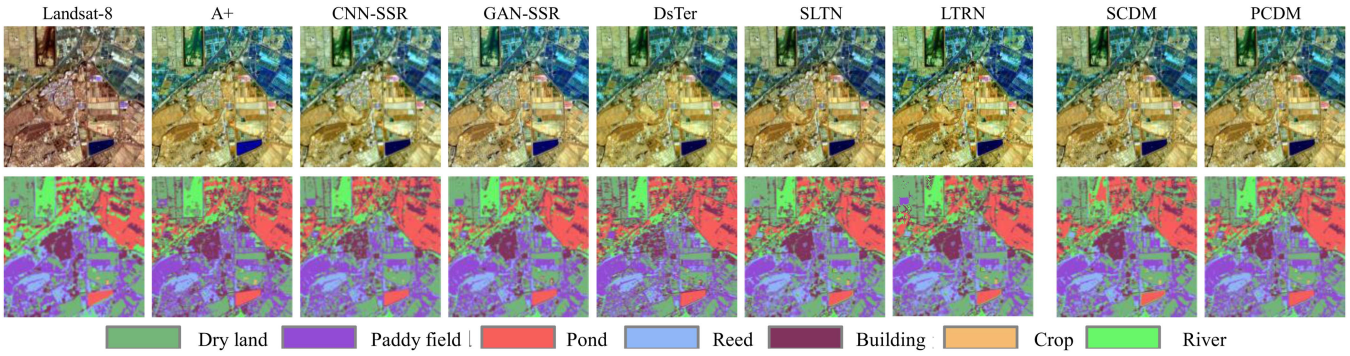


Fig. 8. Spectral reconstruction and classification results of real datasets.

ERGAS and SAM. DsTer, SLTN, LRTN, and SCDM deliver robust results, while PCDM consistently outperforms all methods.

Fig. 7 illustrates band-wise SSIM values. A+ demonstrates the poorest SSIM across nearly all bands in all datasets. CNN-SSR and GAN-SSR exhibit high inter-band variability. DsTer, SLTN, LRTN, and SCDM maintain stable SSIM values, and PCDM achieves the highest SSIM in nearly every band.

D. Spectral Super-Resolution With Real-World

To further evaluate the practical application performance of PCDM, we collected available Landsat-PRISMA image pairs for training the network and used Landsat data to perform spectral reconstruction of the corresponding PRISMA HSI. Fig. 8 presents the spectral reconstruction results of different methods on the real-world dataset. All methods preserve the spatial detail of MSI. However, A+, CNN-SSR, and GAN-SSR exhibit color discrepancies in the water regions compared to other methods, DsTer, SLTN, LRTN, and SCDM exhibit relatively consistent visualization results, while PCDM shows consistent color distribution with the other benchmark methods, and its color representation in both water and dry regions aligns better with the actual scenario. In addition, PCDM demonstrates clearer spatial details compared to the other methods.

The second row of Fig. 8 presents the classification performance of spectral super-resolution results using different methods. In this experiment, we employed a support vector machine as the classifier. The training and validation samples were

TABLE II
CLASSIFICATION ACCURACY EVALUATION RESULTS

Methods	OA (%)	Kappa
Landsat-8	79.41	0.756
A+	81.69	0.789
CNN-SSR	85.26	0.833
GAN-SSR	85.95	0.846
DsTer	88.60	0.831
SLTN	87.97	0.828
LRTN	88.72	0.817
SCDM	90.31	0.869
PCDM	92.19	0.883

visually selected from high spatial resolution imagery on Google Earth, with a 7:3 ratio. The overall classification accuracy (OA) and Kappa coefficient were used to evaluate the classification results. The experimental results show that the original Landsat image struggles to effectively differentiate water bodies and pits, while CNN-SSR and GAN-SSR have misclassifications along object boundaries. DsTer, SLTN, LRTN, SCDM, and PCDM produce more consistent classification results, although some differences still exist in the reed areas. Table II displays the quantitative evaluation accuracy of different classification results. All methods outperform the Landsat-8 image in classification accuracy, indicating that spectral reconstruction can effectively enhance target recognition accuracy. A+ achieves the worst classification results among all methods, with an OA improvement of 2.28 and a Kappa increase of 0.033. CNN-SSR and GAN-SSR have similar classification accuracies, with OA

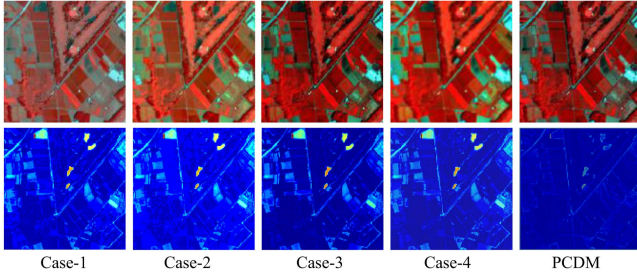


Fig. 9. Comparison results of ablation experiments.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENT

Case	IRM	CSM	FFM	SAM	PSNR
Case-1	×	√	√	7.23	28.36
Case-2	×	×	√	12.98	25.73
Case-3	√	√	×	4.33	33.66
Case-4	One stage			6.69	29.75
	PCDM			1.98	38.43

around 85.5 and Kappa around 0.84. DsTer, SLTN, and LRTN achieve better classification accuracy, with OA improving by around 8 and Kappa by 0.07 compared to the Landsat image. SCDM obtains the highest OA and Kappa among the benchmark methods, and our proposed PCDM outperformed all other methods in classification accuracy.

E. Discussion

Compared to the state-of-the-art methods, PCDM achieves the best spectral reconstruction results on both simulated and real datasets. This is primarily due to the following reasons.

- 1) PCDM constructs a multistage model for progressive spectral reconstruction, which achieves optimal reconstruction results by minimizing the error at each stage.
- 2) PCDM integrates the CSM and IRM modules with the multistage CDM in a coordinated manner, where the interdependencies between network modules ensure the rationality of the network structure.
- 3) In the CSM and IRM, we design modules for extracting both local and global information from the image, effectively leveraging the rich feature information of the image, which facilitates subsequent channel synthesis and image reconstruction. Furthermore, we develop a multi-scale FFM, which effectively integrates the outputs from multiple stages to achieve the final high-precision spectral reconstruction.

To further validate the contributions of different subnetwork modules to PCDM, we constructed ablation experiments with four cases, using the Chikusei simulated dataset for evaluation. In Case 1, we ablated the IRM module. As shown in Fig. 9, the spectral super-resolution results of Case 1 exhibit significant spectral distortion. Table III shows that the SAM increased by 5.25 and the PSNR decreased by 10.07, indicating that the IRM module is crucial for PCDM and can effectively enhance spectral reconstruction performance. In Case 2, we ablated both the IRM and CSM modules. The ablated results show global

spectral reconstruction errors, with the SAM increasing by 11 and the PSNR decreasing by 12.7. The experimental results demonstrate that the combination of CSM and IRM significantly improves spectral reconstruction accuracy, making them essential for PCDM. In Case 3, we ablated the FFM and replaced it with an additive fusion method for combining features from different stages. As shown in Fig. 9, when the FFM is ablated, the spectral super-resolution results show significant spectral distortion. Table III indicates that the SAM increased by 2.35 and the PSNR decreased by 4.77, which suggests that the FFM design effectively integrates multistage features and contributes to the performance improvement of PCDM. Finally, we conducted an experiment using only a single-stage network to assess the contribution of the multistage setting to PCDM. The results show that using only one stage does not allow the network to converge within a limited number of epochs. The spectral super-resolution results exhibit significant spectral distortion, with the SAM increasing by 4.71 and the PSNR decreasing by 8.68.

We also conducted a statistical analysis of the SSIM values for different methods at each epoch to examine the convergence behavior of the methods, as shown in Fig. 10. In the PaviaC dataset, the SSIM values of CNN-SSR and GAN-SSR continued to rise over 200 epochs but did not converge, while DsTer, SLTN, LRTN, and SCDM reach their peak SSIM values at epoch 150 and remained steady between epochs 150 and 200. The proposed PCDM shows a rapid increase in SSIM from epoch 1 to 60, reaching its peak, and then maintains stability between epochs 60 and 200. In the Xiong'an dataset, CNN-SSR and GAN-SSR exhibit a rapid increase in SSIM from epochs 1 to 100, followed by a slower increase from epoch 101 to 160, reaching their peak and stabilizing between epochs 160 and 200. DsTer, SLTN, LRTN, and SCDM show rapid increases in SSIM from epochs 1 to 50 and then stabilized. PCDM continuously optimizes network parameters from epochs 1 to 40 and maintains stable performance from epochs 40 to 200. For the Chikusei dataset, CNN-SSR and GAN-SSR fail to converge within 200 epochs, while DsTer, SLTN, LRTN, and SCDM reach their peak SSIM values at epoch 150. PCDM shows a rapid performance improvement and reached its peak within 100 epochs. Therefore, the simulation experiments demonstrate that PCDM achieves faster convergence rates, outperforming state-of-the-art methods.

In addition, a comparative analysis of computational complexity across these methods is summarized in Table IV. The computational complexity analysis demonstrates distinct characteristics across methods: A+ (implemented in MATLAB) exhibits significantly higher inference time (496.67 s) compared to deep learning counterparts, though parameter and FLOPs metrics are unavailable. CNN-SSR (7.78M parameters, 5.23G FLOPs) achieves the fastest execution (3.03 s) among Python-based methods, while GAN-SSR (9.55M, 6.56G) marginally increases computational demands (3.21 s). DsTer (13.33M, 8.06G) and SLTN (12.56M, 7.89G) show the heaviest model architectures, corresponding to prolonged inference times (5.09 s and 5.26 s, respectively). LRTN (10.44M, 6.72G) and SCDM (9.65M, 8.05G) balance parameters and FLOPs with

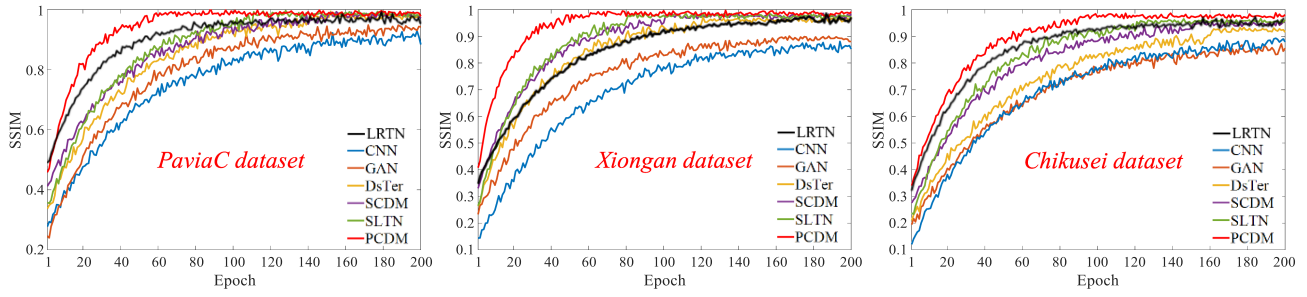


Fig. 10. Comparison results of convergence rates of different network.

TABLE IV
COMPARATIVE ANALYSIS OF COMPUTATIONAL COMPLEXITY

Running information	A+	CNN-SSR	GAN-SSR	DsTer	SLTN	LTRN	SCDM	PCDM
Param (M)	—	7.78	9.55	13.33	12.56	10.44	9.65	11.24
FLOPS (G)	—	5.23	6.56	8.06	7.89	6.72	8.05	6.78
Time (s)	496.67	3.03	3.21	5.09	5.26	4.67	3.97	4.12
Software Platform	Matlab	Python	Python	Python	Python	Python	Python	Python

moderate runtime (3.97 s and 4.67 s, respectively). Our PCDM method (11.24M parameters) maintains competitive efficiency with 6.78G FLOPs and 4.12 s inference time, outperforming DsTer and SLTN in computational load while retaining practical applicability within Python frameworks.

V. CONCLUSION

To impose constraints on the intermediate processes of the network and achieve a global optimal solution through local error minimization, we propose the PCDM for multistage spectral restoration of remote sensing image. PCDM constructs a CSM to generate GT set, and IRM ensures the validity of GT set. The parameters of the IRM are shared with multispectral images to obtain the conditional matrix set. In addition, four conditional diffusion models are constructed, with GT set and the conditional matrix set input separately. The outputs of different stages are then fused through an FFM to obtain the final spectral reconstruction result. Through experiments conducted on both simulated and real datasets, and by comparing with six state-of-the-art methods, we draw the following conclusions.

- 1) PCDM achieves the best results on both simulated and real datasets, outperforming the advanced methods, and can effectively reconstruct spectral information while preserving spatial details.
- 2) The network design of PCDM is rational, and the sub-network modules are critical for improving the network performance.
- 3) Compared to existing deep learning algorithms, PCDM exhibits a faster convergence rate and can obtain the optimal solution within a limited number of epochs.

However, the extensive use of network modules in PCDM leads to suboptimal computational complexity. In future work, we will integrate knowledge distillation theory with PCDM to enhance computational efficiency while maintaining spectral super-resolution performance.

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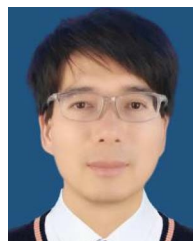
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